

Deploying rietveld-agent on OpenCode, Claude Code, and Codex: installation, verification, and governance

Paper P-3 of the paper series of the rietveld-agent repository

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Abstract

Thesis. The scientific engine of `rietveld-agent` is runtime-agnostic: the same deterministic pipeline (Paper P-1) and the same validation evidence (Paper P-2) are available from any agent-aware terminal — OpenCode, Claude Code, or Codex — on macOS, Linux, or Windows (native or WSL), with a three-command installation, scripted dependency bootstrapping (GSAS-II pinned, never bundled), and a checkable verification path (`make check`, `make env`). This paper gives the installation matrix for the three runtimes and the three platforms, the first-run verification, and the governance the repository imposes on agent-driven scientific work (versioned, hashed scientific inputs; no telemetry; no cloud refinement; interchangeable hosts around one engine). Composition follows [1].

1 Introduction

A repository is only as portable as its installation story. Scientific repositories have an additional requirement: whoever runs them — human or agent — must be able to verify that what they got is what was measured. *Gap.* Agent runtimes (OpenCode, Claude Code, Codex) are terminal-native, but most scientific repositories assume interactive GUI sessions, ad-hoc environments, or cloud execution. *Response.* The `rietveld-agent` repository [2] is built for non-interactive execution from the start: every result is rebuilt by `make` targets, GSAS-II [3] is pinned in `.vendor/` and bootstrapped deterministically, and an `AGENTS.md` orientation file points any agent at the contracts it must respect. This paper documents installation, verification, and governance for the three runtimes.

2 Installation matrix

Requirements in all cases: Python 3.10+ and git. GSAS-II (Apache-2.0) is pinned in `.vendor/GSAS-II` and installed by `benchmarks/eval/sim.py:ensure_gsasii` on first use; it is never bundled as a dependency. `tectonic` is required only for building the paper PDFs (`make paper`), not for running analyses.

2.1 OpenCode

platform	install & verify
macOS	<code>brew install opencode</code> or <code>npm i -g opencode-ai</code> ;
Linux	<code>curl -fsSL https://opencode.ai/install bash</code> or <code>npm i -g opencode-ai</code> ;
Windows	<code>npm i -g opencode-ai</code> (native), or WSL + the Linux route below.

Verify with `opencode -version`, then start the agent in the repository root: `cd rietveld-agent && opencode`.

2.2 Claude Code

platform	install & verify
macOS / Linux	<code>npm install -g @anthropic-ai/claude-code</code> (or the native installer script);
Windows	native installer, or WSL + the Linux route.

Authenticate with `claude` (browser login) or an `ANTHROPIC_API_KEY`; verify with `claude -version`; start with `cd rietveld-agent && claude`.

2.3 Codex (OpenAI)

platform	install & verify
macOS	<code>brew install codex</code> or <code>npm i -g @openai/codex</code> ;
Linux	<code>curl -fsSL https://codex.openai.com/install.sh bash</code> or <code>npm i -g @openai/codex</code> ;
Windows	WSL + the Linux route (codex is a terminal-native CLI).

Authenticate with `codex login` (or `OPENAI_API_KEY`); verify with `codex -version`; start with `cd rietveld-agent && codex`.

3 First run and verification

```
make env      # virtualenv with numpy/scipy/matplotlib
make check    # syntax + structure checks (fast, no network)
make report   # rerun every RQPA refinement (GSAS-II, ~45 min)
make paper    # rebuild all paper PDFs (paper/*.tex, tectonic)
```

The full validation evidence is reproduced with `python3 benchmarks/spikes/spike_16_validate.py` (~32 min), or in parts with `-skip-rerun` (known-answer + gates only) and `-skip-synth` (rerun + gates only). Any agent can therefore verify the engine from scratch without human intervention, and the canonical content hash of the report (Paper P-2) makes the verification objective.

4 Governance for agent-driven scientific work

The repository enforces five invariants that make agent operation safe: (1) *controlled scientific inputs* — catalog, calibrations, policies, models — are hashed, versioned and reviewable; no update may silently change the meaning of an existing analysis; (2) *local first* — no telemetry, no cloud refinement, Apache-2.0; (3) *weak context only* — the sample name may add candidate

phases, never remove them or improve scores; (4) *defensible uncertainty* — per-phase statuses (supported, inconclusive, not_selected, out_of_domain, held, failed) instead of unearned accuracy claims; and (5) *interchangeable hosts* — the three runtimes above are thin front-ends around the same engine, so results do not depend on which agent drove them. `AGENTS.md` is the single orientation point for any runtime.

5 Closing

Installation, verification, and governance together make `rietveld-agent` usable by humans and agents alike on all major platforms, with objective reproducibility as the anchor ([2]; context: Papers P-1 and P-2 of this series; writing framework [1]). The one external requirement is Python 3.10+; everything else is pinned or scripted.

License

Apache-2.0 for original code; GSAS-II under its own Apache-2.0 license; catalog and structure files CC0 from COD (attribution preserved).

References

- [1] J. M. Drake and B. A. Han. How to write a scientific paper in fifteen steps. *PLoS Computational Biology*, 21(9):e1013505, 2025. doi: 10.1371/journal.pcbi.1013505. PMCID: PMC12459795.
- [2] qaemu. rietveld-agent: a deterministic rietveld QPA engine for scientific agent runtimes, 2026. URL <https://github.com/qaemu/rietveld-agent-spikes>. Apache-2.0; see docs/papers (series P-1..P-3).
- [3] B. H. Toby and R. B. Von Dreele. GSAS-II: the genesis of a modern open-source all purpose crystallography software package. *Journal of Applied Crystallography*, 46:544–549, 2013. doi: 10.1107/S0021889813003531.